

第九章 统计力学理论.

§ 1 经典统计系统.

自由度为 s 的系统, 微观状态:

$$q_1 \cdots q_s; p_1 \cdots p_s$$

其演化满足

$$\begin{cases} \dot{q}_i = \frac{\partial H}{\partial p_i} \\ \dot{p}_i = -\frac{\partial H}{\partial q_i} \end{cases} \quad H = H(q, p) \text{ 为系统 Hamilton 量.}$$

定义相空间积元: $d\Omega = dq_1 \cdots dq_s dp_1 \cdots dp_s$

t 时刻系统处在状态 $(q_1 \cdots q_s; p_1 \cdots p_s)$ 的概率:

$$\rho(q, p, t) d\Omega, \quad \rho \text{ 为几率密度.}$$

$$\text{归一化条件: } \int \rho(q, p, t) d\Omega = 1$$

力学量 O 的宏观观测值

$$O_{\text{macro}} = \bar{O} = \int O(q, p) \rho(q, p, t) d\Omega$$

定义统计系集:

$$\text{系统所有可能微观状态数为 } \mathcal{N} = \int \bar{\rho} d\Omega$$

$$\text{即 } \rho d\Omega = \frac{\bar{\rho} d\Omega}{\int \bar{\rho} d\Omega} \propto \bar{\rho} d\Omega \quad \text{微观状态处于 } d\Omega \text{ 内的数目.}$$

一个微观状态 $\Leftrightarrow$ 一个处于该微观状态下的系统.

$\hookrightarrow$ 统计系集: $\mathcal{N}$ 个系统.

§ 2 Liouville 定理.

对密度为 $\hat{\rho}(q, p, t)$ 的一系代表点:

考虑演化 $t \rightarrow t + dt$.

$$q_i \rightarrow q_i + \dot{q}_i dt, \quad p_i \rightarrow p_i + \dot{p}_i dt.$$

$$\hat{\rho} \rightarrow \hat{\rho}(q_i + \dot{q}_i dt, p_i + \dot{p}_i dt, t + dt)$$

$$\begin{aligned} d\hat{\rho} &= \frac{d\hat{\rho}}{dt} \cdot dt \\ &= \hat{\rho}(q', p', t') - \hat{\rho}(q, p, t) \\ &= \frac{\partial \hat{\rho}}{\partial q} dq + \frac{\partial \hat{\rho}}{\partial p} dp + \frac{\partial \hat{\rho}}{\partial t} dt \\ &= \left(\frac{\partial \hat{\rho}}{\partial q} \dot{q} + \frac{\partial \hat{\rho}}{\partial p} \dot{p} + \frac{\partial \hat{\rho}}{\partial t} \right) dt \end{aligned}$$

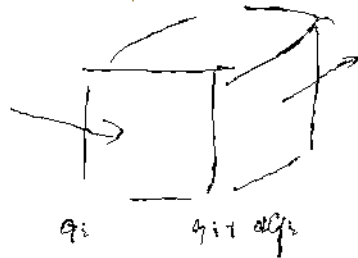
$$\Rightarrow \frac{d\hat{\rho}}{dt} = \frac{\partial \hat{\rho}}{\partial t} + \sum_i \left(\frac{\partial \hat{\rho}}{\partial q_i} \dot{q}_i + \frac{\partial \hat{\rho}}{\partial p_i} \dot{p}_i \right)$$

Liouville 定理, $\frac{d\hat{\rho}}{dt} = 0$, 系统的所有密度在运动中不变.

$$\frac{d\hat{\rho}}{dt} = \frac{d\hat{\rho}}{dt} \cdot dt, \text{ 或, 流函数代表点数.}$$

对于平面流 $q_i \sim q_i + dq_i$

$$\begin{aligned} dN_{q_i} &= dN_{q_{in}} - dN_{q_{out}} \\ &= \hat{\rho}(q_i) |_{q_i} dt d\Omega - \hat{\rho}(q_i) |_{q_i + dq_i} dt d\Omega \\ &= - \frac{\partial(\hat{\rho} q_i)}{\partial q_i} dq_i dt d\Omega = - \frac{\partial(\hat{\rho} q_i)}{\partial q_i} dt d\Omega. \end{aligned}$$



$$\frac{d\hat{\rho}}{dt} = - \sum_i \left[\frac{\partial(\hat{\rho} q_i)}{\partial q_i} + \frac{\partial(\hat{\rho} p_i)}{\partial p_i} \right] = - \nabla \cdot (\hat{\rho} \vec{v})$$

$$\vec{v} = (\dot{q}_1 - \dot{q}_2, \dot{p}_1 - \dot{p}_2)$$

$$\vec{v} = \left(\frac{\partial}{\partial q_1}, \dots, \frac{\partial}{\partial q_s}; \frac{\partial}{\partial p_1}, \dots, \frac{\partial}{\partial p_s} \right)$$

连续性方程: $\frac{d\hat{\rho}}{dt} + \nabla \cdot (\hat{\rho} \vec{v}) = 0$

$$\begin{aligned}
 \hookrightarrow \frac{d\vec{p}}{dt} &= - \sum_i \left[\frac{\partial \vec{p}_i}{\partial q_i} + \frac{\partial \vec{p}_i}{\partial p_i} \right] + \sum_i \left(\frac{\partial \vec{p}}{\partial q_i} q_i + \frac{\partial \vec{p}}{\partial p_i} p_i \right) \\
 &= - \vec{p} \sum_i \left(\frac{\partial q_i}{\partial q_i} + \frac{\partial p_i}{\partial p_i} \right) \\
 &= - \vec{p} \sum_i \left(\frac{\partial}{\partial q_i} \frac{dH}{dp_i} - \frac{\partial}{\partial p_i} \frac{dH}{dq_i} \right) = 0 \quad \square.
 \end{aligned}$$

排他.

$$\begin{aligned}
 \frac{d\vec{p}}{dt} + \sum_i \left(\frac{\partial \vec{p}}{\partial q_i} \frac{dH}{dp_i} - \frac{\partial \vec{p}}{\partial p_i} \frac{dH}{dq_i} \right) \\
 = \frac{d\vec{p}}{dt} + \vec{p} \cdot \nabla H = 0
 \end{aligned}$$

成条件: $\frac{dH}{dt} = 0$ (保守), 系统不受外界干扰

§ 3 微正则系综. 如何求出 ρ ?

1. 考虑处于平衡态的孤立系, $\frac{dH}{dt} = 0$ (Liouville 定理)

$$\frac{dH}{dt} = 0 \quad (\text{平衡}) \Rightarrow \bar{H} = \langle H \rangle$$

等概率假设.

微正则系综.

$$\rho = \begin{cases} C & E \leq H & \text{or} & E < H < E + \delta E \\ 0 & E > H & \text{or} & E + \delta E < H \end{cases} \quad (\delta E \ll E)$$

归一化条件: $\lim_{\delta E \rightarrow 0} \int d\Omega \rho = \lim_{\delta E \rightarrow 0} C \cdot \int_{\delta E} d\Omega = 1$

力学量 O 的平均值: $\bar{O} = \lim_{\delta E \rightarrow 0} C \cdot \int_{\delta E} O d\Omega$

2. 1) ergodic hypothesis.

对于孤立的保守力学系统, 只要时间够长, 从任一初态出发, 都将经过能量曲面上的一切微观状态. X 不成立.

$\hookrightarrow$ quasi-ergodic hypothesis.

... 未经历的区域则为 0 X 不成立.

2) 真实系统为微扰 $\rightarrow$ 各微观状态都会等概率出现.

$\langle \dots \rangle = \frac{1}{\Omega} \int \dots \Omega(E) dE$ $\langle \dots \rangle = \frac{1}{\Omega} \int \dots \Omega(E) dE$ $\langle \dots \rangle = \frac{1}{\Omega} \int \dots \Omega(E) dE$

3. 量子统计系综

$$\rho_s = \begin{cases} C & E \leq E_s \leq E + \Delta E \\ 0 & \text{其它} \end{cases} \quad E, V, N \text{ 时的量子态总数.}$$

$$\sum_s \rho_s = C \cdot \frac{\Omega}{\Omega} = C \cdot \Omega(E, V, N) = 1$$

4. 微正则系综

$$\text{熵} \quad S(E, V, N) = k \ln \Omega(E, V, N)$$

例：计算经典极限下单原子分子组成的理想气体的热力学性质。

$$\Omega(E, V, N) = \frac{1}{N! h^{3N}} \int_{E \leq H \leq E + \Delta E} d\Omega$$

$$\text{其中 } d\Omega = dq_1 \dots dq_{3N} dp_1 \dots dp_{3N}$$

$$H = \sum_{i=1}^N \frac{p_i^2}{2m}$$

$$\Omega(E, V, N) = \frac{1}{N! h^{3N}} \int_{H \leq E} d\Omega$$

$$= \frac{V^N}{N! h^{3N}} \int_{\sum_{i=1}^N \frac{p_i^2}{2m} \leq E} dp_1 \dots dp_{3N} \quad \left| \int dq_1 \dots dq_{3N} = V^N \right|$$

$$= \frac{V^N (2\pi m E)^{3N/2}}{N! h^{3N}} \int dx_1 \dots dx_{3N} \quad x_i = \frac{p_i}{\sqrt{2mE}}$$

$$= K \cdot \frac{V^N (2\pi m E)^{3N/2}}{N! h^{3N}} \quad K \text{ 为 } 3N \text{ 维球的体积.}$$

$$= \frac{V^N (2\pi m E)^{3N/2}}{N! h^{3N} \Gamma(\frac{3N}{2} + 1)}$$

收敛因子 $\delta = 0$

$$\ast \text{ 计算: } \frac{1}{N! h^{3N}} \int e^{-\beta H} d\Omega$$

$$\Omega = \int_0^\infty e^{-\beta E} \Omega(E) dE$$

$$= \int_0^\infty e^{-\beta E} K \cdot \frac{V^N (2\pi m E)^{3N/2}}{N! h^{3N}} \cdot \frac{3N}{2E} dE$$

$$= \frac{3}{2} N K \frac{V^N (2\pi m)^{3N/2}}{h^{3N}} \cdot \int_0^\infty e^{-\beta E} E^{\frac{3N}{2}-1} dE$$

$$\begin{aligned}
 &= \frac{3}{2} N k \frac{V^N (2m)^{3N/2}}{N! h^{3N}} \frac{2}{3N} \left(e^{-\beta E} E^{3N/2} \Big|_0^\infty + \beta \int_0^\infty e^{-\beta E} E^{3N/2-1} dE \right) \\
 &= k \cdot \frac{V^N (2m)^{3N/2}}{N! h^{3N}} \cdot \beta \cdot \frac{3N}{2} \int_0^\infty e^{-\beta E} (E)^{3N/2-1} dE \\
 &= k \frac{V^N (2m)^{3N/2}}{N! h^{3N}} \int (3N/2 - 1)
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{3} \quad & \frac{1}{N! h^{3N}} \int e^{-\beta \vec{p}^2} d\vec{p} \\
 &= \frac{V^N}{N! h^{3N}} \int e^{-\beta \frac{p^2}{2m}} dp_1 \dots dp_{3N} \\
 &= \frac{V^N}{N! h^{3N}} \left(\int e^{-\frac{\beta p^2}{2m}} dp \right)^{3N} = \frac{V^N (2m/\beta)^{3N/2}}{N! h^{3N}} (\pi)^{3N/2}
 \end{aligned}$$

$$\textcircled{1}, \textcircled{2} \rightarrow k = \frac{\lambda^{3N/2}}{\int (3N/2 - 1)}$$

$$\Rightarrow \bar{Z}(E, V, N) = \frac{V^N (2\lambda m E)^{3N/2}}{N! h^{3N} (3N/2)!}$$

$$\bar{Z}'(E, V, N) = \frac{\partial}{\partial E} \bar{Z}(E, V, N) = \frac{V^N (2\lambda m E)^{3N/2}}{N! h^{3N} (3N/2)!} \left(\frac{3N}{2} \right) \frac{1}{E}$$

$$\Omega(E, V, N) = \bar{Z}(E + \Delta E, V, N) - \bar{Z}(E, V, N)$$

$$= \bar{Z}'(E, V, N) \Delta E$$

$$= \frac{V^N (2\lambda m E)^{3N/2}}{N! h^{3N} (3N/2)!} \left(\frac{3N}{2} \right) \frac{\Delta E}{E}$$

$$\begin{aligned}
 \ln \Omega(E, V, N) &= N \ln V + \frac{3N}{2} \ln(2\lambda m E) - \ln(N!) - 3N \ln h \\
 &\quad - \ln \left(\frac{3N}{2}! \right) + \ln \left(\frac{3N}{2} \right) + \ln \frac{\Delta E}{E}
 \end{aligned}$$

$$\approx N \ln V + \frac{3N}{2} \left[\ln(2\lambda m E) - 2 \ln N \right] - N (\ln N - 1) - \frac{3N}{2} (\ln \frac{3N}{2} - 1) + \ln \frac{3N}{2} + \ln \frac{\Delta E}{E}$$

$$= N \ln \frac{V}{N} + \frac{3N}{2} \ln \left[\frac{2\lambda m E}{h^2} \cdot \frac{2}{3N} \right] + \frac{5}{2} N + \ln \frac{3N}{2} + \ln \frac{\Delta E}{E}$$

$$\approx N \ln \frac{V}{N} + \frac{3N}{2} \ln \left(\frac{4\lambda m E}{3h^2 N} \right) + \frac{5}{2} N$$

$$\langle E, V, N \rangle = k \left[\ln \frac{V}{N} + \ln \left(\frac{4\lambda m E}{3h^2 N} \right)^{3/2} + \frac{5}{2} \right]$$

$$dS = \frac{1}{T} dU + \frac{P}{T} dV - \frac{\mu}{T} dN$$

$$\left(\frac{\partial S}{\partial E}\right)_{V,N} = \frac{1}{T} = Nk \cdot \frac{3}{2} \cdot \frac{1}{E} \Rightarrow E = \frac{3}{2} NkT$$

$$\left(\frac{\partial S}{\partial V}\right)_{E,N} = \frac{P}{T} = Nk \frac{1}{V} \Rightarrow PV = NkT$$

$$\left(\frac{\partial S}{\partial N}\right)_{E,V} = -\frac{\mu}{T} = -\frac{3}{2} k \ln N + k \ln V + k \ln \left(\frac{2\pi m E}{3h^2}\right)^{3/2}$$

$$\begin{aligned} \Rightarrow \mu &= kT \left[\frac{3}{2} \ln N - \ln V - \frac{3}{2} \ln \left(\frac{2\pi m NkT}{h^2} \right) \right] \\ &= kT \left[\ln \frac{N}{V} + \ln \left(\frac{h^2}{2\pi m kT} \right)^{3/2} \right] \end{aligned}$$

$$S(T, V, N) = Nk \ln \left(\frac{V}{N} \right) + Nk \ln \left(\frac{2\pi m kT}{h^2} \right)^{3/2} + \frac{5}{2} Nk$$

$$N \gg 1 \text{ 时, } \ln \tilde{z} \sim m \tilde{z}' \sim \ln \Omega$$

$$S = k \ln \tilde{z} \approx k \ln \tilde{z}' = k \ln \Omega$$

§ 4 正则系综

温度 T 确定

1. 宏观条件: 系统与 大热源 / 热库 接触达到平衡。

$\Rightarrow T, V, N$ 一定。

系统 + 热库: E 一定, 微正则系综。

$$E = E_1 + E_2$$

令 $p_{12}(E_1)$ 为系统处于能量为 E_1 的状态的概率。

$$\text{则 } p_{12}(E_1) = \frac{\Omega_1(E - E_1)}{\Omega(E)}$$

E_1 时, Ω_1 为所有可取的 E_1 的状态数。
 $E_1 + E_2$ 时, Ω_2 为所有可取的 E_2 的状态数。

$$“大”热库 $\rightarrow E_1 \ll E_2 \sim E$$$

$$\text{其中 } \Omega_2(E - E_1) \sim (E - E_1)^m, m \sim O(N)$$

取反对数后展开:

$$\langle E - E_1 \rangle^m = e^{m \ln(E - E_1)} \approx e^{m \ln E} \left(1 - \frac{E_1}{E} + \frac{1}{2} \left(\frac{E_1}{E} \right)^2 \dots \right)$$

$$p_s(E_1) = \frac{1}{\Omega(E)} e^{\ln[\Omega_s(E - E_1)]} \approx \frac{1}{\Omega(E)} e^{\ln \Omega_s(E) - \frac{\partial \ln \Omega_s(E)}{\partial E} E_1}$$

$$= \frac{\Omega_s(E)}{\Omega(E)} e^{-\beta E_1} \quad (\beta = \frac{\partial \ln \Omega_s(E)}{\partial E} \text{ 由大热源决定, } \beta = 1/(kT))$$

$$= \frac{1}{Z_N} e^{-\beta E_1} \quad Z_N = \frac{\Omega(E)}{\Omega_s(E)} = \sum_s e^{-\beta E_s}$$

2. 计算热力学量

$$\text{内能: } \bar{E} = \sum_s E_s p_s = \frac{1}{Z_N} \sum_s E_s e^{-\beta E_s}$$

$$= \frac{1}{Z_N} \left(- \frac{\partial}{\partial \beta} Z_N \right) = - \frac{\partial}{\partial \beta} \ln Z_N$$

$$\text{压强: } \bar{p} = \sum_s \frac{\partial E_s}{\partial y_1} p_s = \frac{1}{Z_N} \sum_s \frac{\partial E_s}{\partial y_1} e^{-\beta E_s}$$

$$= \frac{1}{Z_N} \left(\frac{1}{\beta} \right) \frac{\partial}{\partial y_1} \sum_s e^{-\beta E_s} = - \frac{1}{\beta} \frac{\partial}{\partial y_1} \ln Z_N$$

$$P = \frac{1}{\beta} \frac{\partial}{\partial y_1} \ln Z_N$$

$$\text{熵: } S = \frac{1}{T} \partial Q = \frac{1}{T} (dU - \partial W) = \frac{1}{T} (dU - \gamma dy_1)$$

$$= \frac{1}{T} \left[d \left(- \frac{\partial \ln Z_N}{\partial \beta} \right) + \frac{1}{\beta} \sum_i \left(\frac{\partial}{\partial y_i} \ln Z_N dy_i \right) \right]$$

$$d[\ln Z(\beta, y)] = \frac{\partial \ln Z}{\partial \beta} d\beta + \sum_i \frac{\partial \ln Z}{\partial y_i} dy_i$$

$$= \frac{1}{T} \left[d \left(- \frac{\partial \ln Z_N}{\partial \beta} \right) + \frac{1}{\beta} \left(d \ln Z_N - \frac{\partial \ln Z_N}{\partial \beta} d\beta \right) \right]$$

$$= \frac{1}{\beta T} \left\{ d \ln Z_N - \beta d \left(\frac{\partial \ln Z_N}{\partial \beta} \right) + \frac{\partial \ln Z_N}{\partial \beta} d\beta \right\}$$

$$= \frac{1}{\beta T} d \left(\ln Z_N - \beta \frac{\partial \ln Z_N}{\partial \beta} \right)$$

$$\Rightarrow S = \frac{1}{\beta T} \left[\ln Z_N - \beta \frac{\partial \ln Z_N}{\partial \beta} \right]$$

$$F = \bar{E} - TS = -\frac{1}{\beta} \ln Z_N = -kT \ln Z_N \quad (\beta = \frac{1}{kT})$$

3. 热力学第零定律、绝对熵。

$$S_0 = \lim_{T \rightarrow 0} S(T, \eta) = 0 \quad \text{是量子效应的宏观体现。}$$

利用正则系综证明：

$$Z_N = \sum_n g_n e^{-\beta E_n}, \quad g_n \text{ 为 } n \text{ 能级的简并度。}$$

$$F = -kT \ln Z_N \xrightarrow{T \rightarrow 0} -kT \ln g_0 e^{-\beta E_0} = E_0 - kT \ln g_0$$

$$S = \frac{1}{T} (\bar{E} - F) = k \ln g_0 = \begin{cases} 0 & (g_0 = 1) \\ \ln g_0 & (g_0 > 1) \end{cases}$$

此时 $\frac{S}{N} \rightarrow 0$, 每个粒子的熵为 0。

4. 能量涨落。

$$\overline{(E - \bar{E})^2} = \bar{E}^2 - 2\bar{E}\bar{E} + \bar{E}^2 = \bar{E}^2 - \bar{E}^2$$

$$\bar{E}^2 = \sum_i E_i^2 p_i = \frac{1}{Z_N} \sum_i E_i^2 e^{-\beta E_i} = \frac{1}{Z_N} \frac{\partial^2}{\partial \beta^2} \sum_i e^{-\beta E_i}$$

$$= \frac{1}{Z_N} \frac{\partial^2}{\partial \beta^2} Z_N = \frac{1}{Z_N} \frac{\partial}{\partial \beta} \left(\frac{\partial Z_N}{\partial \beta} \cdot \frac{\partial \ln Z_N}{\partial \beta} \right) = \frac{1}{Z_N} \cdot \frac{\partial}{\partial \beta} \left(\bar{E} Z_N \frac{\partial \ln Z_N}{\partial \beta} \right)$$

$$= \frac{1}{Z_N} \frac{\partial \bar{E}}{\partial \beta} \cdot \frac{\partial \ln Z_N}{\partial \beta} + \frac{\partial^2 \ln Z_N}{\partial \beta^2} = \bar{E}^2 - \frac{\partial \bar{E}}{\partial \beta}$$

$$(E - \bar{E})^2 = -\frac{\partial \bar{E}}{\partial \beta} = kT^2 \left(\frac{\partial \bar{E}}{\partial T} \right)_{V, N} = kT^2 C_V$$

$$\frac{\overline{(E - \bar{E})^2}}{\bar{E}} = \frac{kT^2 C_V}{\bar{E}} \sim \frac{1}{N} = \frac{1}{N} \rightarrow 0, \quad \text{在热力学极限下,}$$

$C_V, \bar{E}$ 为广延量。

$N \rightarrow \infty$

正则/微正则系综求得系统平衡性质相同。

5. 经典极限

$$\lambda_T = \frac{h}{(2\pi m kT)^{1/2}} \ll \bar{r}, \quad \Delta E \ll kT$$

$$= H(q, p)$$

$$\beta(q, p) d\Omega = \frac{1}{Z_N N! h^3} e^{-\beta H} d\Omega, \quad d\Omega = q_1 \dots q_3 p_1 \dots p_3$$

$$Z_N = \frac{1}{N! h^3} \int e^{-\beta H(q, p)} d\Omega$$

6. 例. 经典极限下的单原子分子气体.

$$Z_N = \frac{1}{N! h^{3N}} \int e^{-\beta H} d\Omega$$

$$H = \sum_{i=1}^N \epsilon_i = \sum_{i=1}^N \frac{\vec{p}_i^2}{2m}, \quad d\Omega = \prod_{i=1}^N d\omega_i = \prod_{i=1}^N dx_i dy_i dz_i dp_{xi} dp_{yi} dp_{zi}$$

$$Z_N = \frac{1}{N! h^{3N}} \int e^{-\beta \sum_{i=1}^N \epsilon_i} \prod_{i=1}^N d\omega_i$$

$$= \frac{1}{N!} \prod_{i=1}^N \int \frac{1}{h^3} \cdot e^{-\beta \epsilon_i} d\omega_i$$

$$= \frac{z^N}{N!} \quad z = \frac{V}{h^3} \left(\frac{2\pi m}{\beta} \right)^{3/2} \text{ 平动分子配分函数.}$$

$$\ln Z_N = N \ln z - \ln N!$$

$$\bar{E} = - \frac{\partial}{\partial \beta} \ln Z_N = - N \frac{\partial}{\partial \beta} \ln z = \frac{3}{2} N k T$$

$$P = \frac{1}{\beta} \frac{\partial}{\partial V} \ln Z_N = \frac{1}{\beta} N \frac{\partial}{\partial V} \ln z = \frac{N k T}{V}$$

$$S = k (\ln Z_N - \beta \frac{\partial \ln Z_N}{\partial \beta})$$

$$= k \left[(N \ln z - \ln N!) + \beta \cdot \frac{3}{2} N k T \right]$$

$$\approx k \left\{ N \ln \left[\frac{V}{h^3} \left(\frac{2\pi m}{\beta} \right)^{3/2} \right] - N \ln N + N + \frac{3}{2} N \right\}$$

$$= k N \left[\ln \frac{V}{N} + \frac{5}{2} + \ln \left(\frac{2\pi m k T}{h^2} \right)^{3/2} \right]$$

$$F = - k T \ln Z_N = k T \cdot \{ N \ln \left[V \cdot \left(\frac{2\pi m k T}{h^2} \right)^{3/2} \right] - \ln N! \}$$

从 $F = F(T, V, N)$ 出发, 可以得到

$$P = - \left(\frac{\partial F}{\partial V} \right)_{T, N}$$

$$S = - \frac{\partial F}{\partial T}$$

$$\mu = \left\langle \frac{\partial F}{\partial N} \right\rangle_{v,T}$$

$$F = F + TS = F - T \cdot \left\langle \frac{\partial F}{\partial T} \right\rangle_{v,N} = -T^2 \left[\frac{\partial}{\partial T} \left(\frac{F}{T} \right) \right]_{v,N}$$

7. example: 极端相对论气体.

$$Z_N = \frac{1}{N! h^{3N}} \int e^{-\beta \sum_i c |\mathbf{p}_i|} d\mathbf{q}^3 d\mathbf{p}^3$$

$$= \frac{V^N}{N! h^{3N}} \left(\int e^{-\beta c p} dp \right)^N$$

$$= \frac{1}{N!} \left[\frac{8\pi V k^3}{h^3 c^3} \right]^N$$

$$U = - \frac{\partial}{\partial \beta} \ln Z_N = 3 N k T$$

$$P = \frac{1}{\beta} \frac{\partial}{\partial V} \ln Z_N = \frac{1}{V} N k T$$

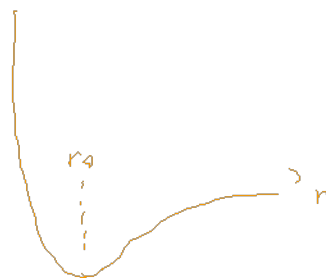
§ 5 非理想气体的物态方程

假设: ① 经典极限条件

② 分子之间是两两相互作用.

③ 只有短程的简单相互作用.

$\phi_{\text{cons}} //$



$$E = K + \phi = \sum_i \frac{\vec{p}_i^2}{2m_i} + \sum_{i < j} \phi_{ij} \quad \phi_{ij} = \phi(|\vec{r}_{ij}|) = \phi(|\vec{r}_i - \vec{r}_j|)$$

配分函数为: $Z_N = \frac{1}{N! h^{3N}} \int e^{-\beta(K+\phi)} d^3\vec{r}_1 \cdots d^3\vec{r}_N d^3\vec{p}_1 \cdots d^3\vec{p}_N$

$$= \frac{1}{N! \lambda_T^{3N}} \int e^{-\beta \sum_{i < j} \phi_{ij}} d^3\vec{r}_1 \cdots d^3\vec{r}_N \quad \lambda_T = \frac{h}{(2\pi m k)^{1/2}}$$

$$= \frac{1}{N! \lambda_T^{3N}} Q_N(\beta, V) \quad \text{处处有限. } \text{若 } \phi_{\text{cons}} \text{ 为升势则 } \phi(r) < 1? \Rightarrow e^{-\beta \phi_{\text{cons}}} \gg 1?$$

令 $f_{ij} = f(r_{ij}) = e^{-\beta \phi_{ij}} - 1$



$$Q_N(\beta, V) = \int \prod_{i < j} (1 + f_{ij}) d^3\vec{r}_1 \cdots d^3\vec{r}_N$$

作“密展展开”

$$= \int \left(1 + \sum_{i < j} f_{ij} + \sum_{i < j} \sum_{i' < j'} f_{ij} f_{i'j'} + \cdots \right) d^3\vec{r}_1 \cdots d^3\vec{r}_N$$

$$\approx \int \left(1 + \sum_{i < j} f_{ij} \right) d^3\vec{r}_1 \cdots d^3\vec{r}_N$$

$$= V^N + \frac{1}{2} N(N-1) \int d^3\vec{r}_3 \cdots d^3\vec{r}_N \int f_{12} d^3\vec{r}_1 d^3\vec{r}_2$$

$$= V^N + \frac{1}{2} N(N-1) V^{N-2} \int d^3\vec{r}_1 \int f(|\vec{r}_1 - \vec{r}_2|) d^3\vec{r}_2$$

规定特征长度 r^* : $r > r^*, f \approx 0$

则积分 $\int f_{12} d^3\vec{r}_2$ 只在 $|\vec{r}_{12}| < r^*$ 范围有值.

取近似 $V \gg r^{*3}$ 忽略边界效应.

$$\approx V^N \left(1 + \frac{N^2}{2V} \int f_{12} d^3 r_{12} \right)$$

$$\ln Q^N = N \ln V + \ln \left(1 + \frac{N^2}{2V} \int f_{12} d^3 \vec{r}_{12} \right) \approx N \ln V + \frac{N^2}{2V} \int f_{12} d^3 \vec{r}_{12}$$

通过配分函数可求出热力学量.

$$P = \frac{1}{\beta} \frac{\partial}{\partial V} \ln Z_N = \frac{1}{\beta} \frac{\partial}{\partial V} \ln Q_N$$

$$= \frac{1}{\beta} \left(\frac{N}{V} - \frac{N^2}{2V^2} \int f_{12} d^3 \vec{r}_{12} \right)$$

级数展开:

$$= \frac{NkT}{V} \left(1 - \frac{N}{2V} \int f_{12} d^3 \vec{r}_{12} \right) \sim P = \frac{NkT}{V} \left(1 + \frac{B_2}{V} + \frac{B_3}{V^2} + \dots \right)$$

$$\hookrightarrow B_2 = -\frac{N}{2} \int f_{12} d^3 \vec{r}_{12}$$

例: 用 ϕ 的几种形式求 B_2 .

Lennard-Jones 势: $\phi(r) = \phi_0 \left\{ \left(\frac{r_0}{r} \right)^{12} - \left(\frac{r_0}{r} \right)^6 \right\}$

带吸引力的刚球势: $\phi(r) = \begin{cases} +\infty & r < r_0 \\ -\phi_0 \left(\frac{r_0}{r} \right)^6 & r \geq r_0 \end{cases}$



例: $B_2 = -\frac{N}{2} \int (e^{-\phi(r)/kT} - 1) d^3 \vec{r}_{12}$

$$= -2\lambda N \int_0^\infty (e^{-\phi(r)/kT} - 1) r^2 dr$$

$$= 2\lambda N \left[\int_0^{r_0} r^2 dr - \int_{r_0}^\infty (e^{-\phi_0 \left(\frac{r_0}{r} \right)^6} - 1) r^2 dr \right]$$

高温展开: $\phi(r)/kT \ll 1 \quad e^{-\phi(r)/kT} \approx 1 - \frac{\phi(r)}{kT}$

...

$$\Rightarrow B_2 = 2\lambda N \left(\frac{r_0^3}{3} - \phi_0 \frac{r_0^3}{3kT} \right) = Nb - \frac{Na}{kT}$$

$$P = \frac{NkT}{V} \left(1 + \frac{Nb}{V} - \frac{Na}{kT} \right) = \frac{NkT}{V} \left(1 + \frac{Nb}{V} - \frac{Z_a}{V} \right)$$

$$\approx NkT \cdot \frac{1}{V - Nb} - \frac{N^2 a}{V}$$

$$\Rightarrow \langle p + \frac{v^2}{V} \rangle (V - Nb) = NkT.$$

§ 6 流体的二粒子分布函数与关联函数.

$$H = K + \phi = \sum_i \frac{\vec{p}_i^2}{2m} + \sum_{i < j} \phi(r_{ij})$$

$$\text{几率密度: } \rho(\vec{r}_1, \dots, \vec{r}_N; \vec{p}_1, \dots, \vec{p}_N) = \frac{1}{N! h^{3N} \Omega_N} e^{-\beta H}$$

↓ 对 $\vec{p}_1, \dots, \vec{p}_N$ 积分.

N 个粒子各有位于 $d^3\vec{r}_1, d^3\vec{r}_2, \dots, d^3\vec{r}_N$ 内的概率:

$$\rho_N(\vec{r}_1, \dots, \vec{r}_N) = \frac{1}{\Omega_N} e^{-\beta \phi} = \frac{e^{-\beta \phi}}{\int e^{-\beta \phi} d^3\vec{r}_1 \dots d^3\vec{r}_N}$$

1. 约化分布函数和关联函数.

定义: 单粒子分布函数 — 粒子 1 在 $d^3\vec{r}_1$ 内的概率.

$$F_1(\vec{r}_1) = N \int \rho_N(\vec{r}_1, \dots, \vec{r}_N) d^3\vec{r}_2 \dots d^3\vec{r}_N$$

$$\text{满足归一化: } \int F_1(\vec{r}_1) d^3\vec{r}_1 = 1$$

对均匀系, 在热力学极限下, 忽略边界效应.

$$F_1 = \frac{N}{V} = n,$$

定义: 二粒子分布函数 — 粒子 1, 2 在 $d^3\vec{r}_1, d^3\vec{r}_2$ 的概率.

$$F_2(\vec{r}_1, \vec{r}_2) = N(N-1) \int \rho_N(\vec{r}_1, \dots, \vec{r}_N) d^3\vec{r}_3 \dots d^3\vec{r}_N$$

$$\text{归一化: } \iint F_2(\vec{r}_1, \vec{r}_2) d^3\vec{r}_1 d^3\vec{r}_2 = N(N-1)$$

对均匀系, 有平移不变性:

$$F_2(\vec{r}_1, \vec{r}_2) = F_2(\vec{r}_1 - \vec{r}_2) = n^2 g(\vec{r}_1 - \vec{r}_2)$$

$$N(N-1) = n^2 \iint g(\vec{r}_1 - \vec{r}_2) d^3\vec{r}_1 d^3\vec{r}_2$$

忽略边界效应.

$$\approx n^2 V \int g(\vec{r}) d^3 \vec{r}$$

$$\hookrightarrow \int g(\vec{r}) d^3 \vec{r} = V(1 - \frac{1}{N}) \approx V$$

极限情况:

$$\textcircled{1} \phi = 0$$

$$F_2 = n^2 (1 - \frac{1}{N}) \approx n^2$$

$$g(\vec{r}) = 1 - \frac{1}{N} \approx 1$$

② 热力学极限, $|\vec{r}| \rightarrow \infty$

$$F_2(\vec{r}) = n^2, \quad g(\vec{r}) = 1$$

引入关联函数 $h(\vec{r}) = g(\vec{r}) - 1$

2. $g(\vec{r})$ 计算内能.

$$\bar{E} = \bar{K} + \bar{\phi}$$

$$= \frac{3}{2} N k T + \frac{1}{2} N(N-1) \overline{\phi(\vec{r}_1 - \vec{r}_2)}$$

$$= \frac{3}{2} N k T + \frac{1}{2} N(N-1) \int \phi(\vec{r}_1 - \vec{r}_2) \rho_N d^3 \vec{r}_1 \dots d^3 \vec{r}_N$$

$$= \frac{3}{2} N k T + \frac{1}{2} \iint \phi_{12}(r_{12}) d^3 \vec{r}_1 d^3 \vec{r}_2 N(N-1) \int \rho_N d^3 \vec{r}_3 \dots d^3 \vec{r}_N$$

$$= \frac{3}{2} N k T + \frac{1}{2} \iint \phi_{12}(r_{12}) F_2(\vec{r}_1, \vec{r}_2) d^3 \vec{r}_1 d^3 \vec{r}_2$$

$$= \frac{3}{2} N k T + \frac{1}{2} n^2 \int d^3 \vec{r}_1 \int d^3 \vec{r}_2 \phi(\vec{r}_2 - \vec{r}_1) g(\vec{r}_2 - \vec{r}_1)$$

短程力, 忽略边界效应.

$$\approx \frac{3}{2} N k T + \frac{N^2}{2V} \int d^3 \vec{r} \phi(r) g(r)$$

$$= \frac{3}{2} N k T + \frac{1}{2} N \cdot \int_0^\infty \phi(r) n g(r) 4\pi r^2 dr$$



3. 用 $g(r)$ 计算压强.

平衡态: 任何量的平均值与时间无关.

$$\frac{d}{dt} \left\langle \sum_{i=1}^{3N} p_i q_i \right\rangle = \overline{\left(\dot{p}_i q_i + q_i \dot{p}_i \right)} = 0$$

$$\overline{q_i \dot{p}_i} = \overline{\frac{\partial H}{\partial p_i} p_i}$$

$$\text{正则系综} = \frac{1}{N! h^{3N} \Omega_N} \int p_i \frac{\partial H}{\partial p_i} e^{-\beta H} d\Omega$$

$$= - \frac{KT}{N! h^{3N} \Omega_N} \int p_i \frac{\partial}{\partial p_i} (e^{-\beta H}) d\Omega$$

$$= \frac{KT}{N! h^{3N} \Omega_N} \left\{ \int e^{-\beta H} d\Omega - \int p_i e^{-\beta H} \Big|_{p_i=-\infty}^{p_i=+\infty} \frac{dp_i}{dp_i} \right\}$$

$$= \frac{KT}{N! h^{3N} \Omega_N} \int e^{-\beta H} d\Omega = KT$$

$$\sum_{i=1}^{3N} \overline{q_i \dot{p}_i} = 3N KT$$

$$\sum_{i=1}^{3N} \overline{p_i \dot{q}_i} = \sum_{i=1}^N \overline{\vec{p}_i \cdot \vec{\dot{r}}_i} = \sum_{i=1}^N \overline{\vec{F}_i \cdot \vec{r}_i}$$

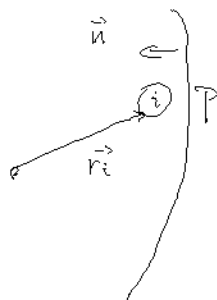
$$\vec{F}_i = \vec{F}_i^{\text{ext}} + \vec{F}_i^{\text{int}}$$

$$= \sum_{i=1}^N \overline{\vec{F}_i^{\text{ext}} \cdot \vec{r}_i} + \sum_{i=1}^N \overline{\vec{F}_i^{\text{int}} \cdot \vec{r}_i}$$

$$= - \int_S p \vec{n} \cdot \vec{r} dA - \sum_i \sum_{j \neq i} \frac{\partial \phi(|\vec{r}_i - \vec{r}_j|)}{\partial \vec{r}_i} \cdot \vec{r}_i$$

$$= - p \int_V (\nabla \cdot \vec{r}) d^3\vec{r} - \sum_{i,j,i \neq j} \frac{\partial \phi(|\vec{r}_i - \vec{r}_j|)}{\partial \vec{r}_i} \cdot \vec{r}_i$$

$$= - 3pV - \frac{1}{2} \sum_{i,j,i \neq j} \langle \vec{r}_i - \vec{r}_j \rangle \frac{\partial \phi(|\vec{r}_i - \vec{r}_j|)}{\partial \vec{r}_i}$$



$$= -3PV - \frac{1}{2} N(N-1) \langle \vec{r}_i - \vec{r}_j \rangle \frac{\partial \phi(|\vec{r}_i - \vec{r}_j|)}{\partial \vec{r}_i}$$

$$= -3PV - \frac{1}{2} \iint \langle \vec{r}_i - \vec{r}_j \rangle \frac{\partial \phi(|\vec{r}_i - \vec{r}_j|)}{\partial \vec{r}_i} \tilde{z}(\vec{r}_i, \vec{r}_j) d^3 \vec{r}_i d^3 \vec{r}_j$$

$$= -3PV - \frac{N^2}{2V} \int |\vec{r}| \cdot \frac{\partial \phi(\vec{r})}{\partial \vec{r}} g(\vec{r}) d^3 \vec{r} = -3NkT$$

$$\Rightarrow P = \frac{NkT}{V} \left(1 - \frac{n}{6kT} \int_0^\infty r \cdot \frac{\partial \phi(r)}{\partial r} g(r) 4\pi r^2 dr \right)$$

4. $g(r)$ 的近似值:

$$\begin{aligned} g(\vec{r}_1, \vec{r}_2) &= \frac{V^2 \int e^{-\phi/kT} d^3 \vec{r}_3 \dots d^3 \vec{r}_N}{\int e^{-\phi/kT} d^3 \vec{r}_1 \dots d^3 \vec{r}_N} \\ &= \frac{V^2 \int e^{-(\phi - \phi_0)/kT} d^3 \vec{r}_3 \dots d^3 \vec{r}_N e^{-\phi_0/kT}}{\int e^{-\phi/kT} d^3 \vec{r}_1 \dots d^3 \vec{r}_N} \\ &= C \cdot e^{-\phi_0/kT} \end{aligned}$$

$$\hookrightarrow g(r) = C \cdot e^{-\phi(r)/kT}$$

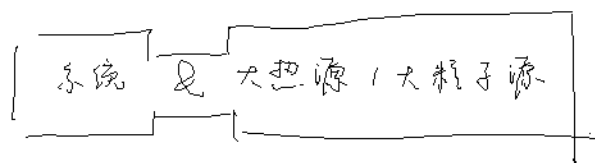
* 结构因子:

$$S(q) = 1 + n \int d^3 \vec{r} g(r) e^{i\vec{q} \cdot \vec{r}}$$

§ 7 正则系综

1. 系统与热源、大粒子源接触达到平衡。

$(T, \mu, \dots)$ - 定。



$$E = E_1 + E_2, \quad N = N_1 + N_2$$

系统(1) 能量为 E_1 , 粒子数为 N_1 时, 处在量 Ω 态 S 的概率,

$$P_{1S}(N_1, E_1) = \frac{\Omega_2(N - N_1, E - E_1)}{\Omega(N, E)}$$

$$\Xi = \frac{\Omega_2(N, E)}{\Omega(N, E)} \quad \text{巨配分函数} \quad \Omega(N, E) e^{\ln \Omega_2(N - N_1, E - E_1)}$$

$$\alpha = \frac{\partial \ln \Omega_2(N, E)}{\partial N} = \frac{-\mu}{kT} \quad \Omega(N, E) e^{\ln \Omega_2(N, E) - \frac{\partial \ln \Omega_2(N, E)}{\partial N} N_1 - \frac{\partial \ln \Omega_2(N, E)}{\partial E} E_1}$$

$$\beta = \frac{\partial \ln \Omega_2(N, E)}{\partial E} = \frac{1}{kT} = \frac{1}{\Xi} e^{-\alpha N_1 - \beta E_1}$$

$\Rightarrow$ 正则系综的概率分布: $P_{NS} = \frac{1}{\Xi} e^{-\alpha N - \beta E}$

$$\text{归一化: } \sum_{N=0}^{\infty} \sum_S P_{NS} = 1$$

2. 计算热力学量。

$$\bar{N} = \sum_{N=0}^{\infty} \sum_S N P_{NS} = \frac{1}{\Xi} \sum_{N=0}^{\infty} \sum_S N e^{-\alpha N - \beta E}$$

$$= -\frac{1}{\Xi} \left(\frac{\partial}{\partial \alpha} \sum_{N=0}^{\infty} \Xi e^{-\alpha N - \beta E} \right) = -\frac{\partial}{\partial \alpha} \ln \Xi$$

$$\bar{E} = \sum_{N=0}^{\infty} \sum_s \bar{E} p_{Ns} = \frac{1}{\Xi} \sum_{N=0}^{\infty} \sum_s \bar{E} \cdot e^{-\alpha N - \beta E} = -\frac{\partial}{\partial \beta} \ln \Xi$$

$$\begin{aligned} \bar{y}_i &= \sum_{N=0}^{\infty} \sum_s \left(\frac{\partial E_s}{\partial y_i} \right) p_{Ns} = \frac{1}{\Xi} \sum_{N=0}^{\infty} \sum_s \frac{\partial E_s}{\partial y_i} e^{-\alpha N - \beta E} \\ &= -\frac{1}{\Xi} \frac{\partial}{\partial y_i} \sum_{N=0}^{\infty} \sum_s \Xi e^{-\alpha N - \beta E} = -\frac{1}{\beta} \frac{\partial}{\partial y_i} \ln \Xi. \end{aligned}$$

$$S = k (\ln \Xi - \alpha \frac{\partial}{\partial \alpha} \ln \Xi - \beta \frac{\partial}{\partial \beta} \ln \Xi)$$

$$\alpha = -\frac{1}{kT}, \quad \beta = \frac{1}{kT}.$$

$$F = \bar{E} - TS = \frac{1}{\beta} \left(\alpha \frac{\partial}{\partial \alpha} \ln \Xi - \ln \Xi \right)$$

$$\bar{\mu} = F - G = F - \bar{N} \mu = -kT \ln \Xi.$$

3. 粒子数 / 能量涨落.

$$\begin{aligned} (1) \quad \overline{(\Delta N)^2} &= \overline{(N - \bar{N})^2} = \bar{N^2} - \bar{N}^2 \\ &= \sum_{N=0}^{\infty} \sum_s N^2 p_{Ns} - \left(\sum_{N=0}^{\infty} \sum_s N p_{Ns} \right)^2 = \frac{1}{\Xi} \frac{d^2}{d\alpha^2} \Xi - \left(\frac{1}{\Xi} \frac{d}{d\alpha} \Xi \right)^2 \\ &= \frac{1}{\Xi} \frac{d}{d\alpha} \left(\frac{d\Xi}{d\alpha} \right) - \left(\frac{1}{\Xi} \frac{d\Xi}{d\alpha} \right)^2 \\ &= \frac{d}{d\alpha} \left(\frac{1}{\Xi} \frac{d\Xi}{d\alpha} \right) - \frac{d}{d\alpha} \left(\frac{1}{\Xi} \right) \frac{d\Xi}{d\alpha} - \left(\frac{1}{\Xi} \frac{d\Xi}{d\alpha} \right)^2 \\ &= -\frac{\partial \bar{N}}{\partial \alpha} + \left(\frac{1}{\Xi} \frac{d\Xi}{d\alpha} \right)^2 - \left(\frac{1}{\Xi} \frac{d\Xi}{d\alpha} \right)^2 \\ &= -\left(\frac{\partial \bar{N}}{\partial \alpha} \right)_{\beta, V} = kT \left(\frac{\partial \bar{N}}{\partial \mu} \right)_{T, V} \quad \alpha = -\frac{\mu}{kT} \end{aligned}$$

$$d\mu = -\frac{S}{N} dT + \frac{V}{N} dp = -sdT + vdp.$$

$$\left(\frac{\partial \mu}{\partial V} \right)_T = v \left(\frac{\partial p}{\partial V} \right)_T.$$

$$\left(\frac{\partial \mu}{\partial N} \right) \cdot \left(\frac{\partial \bar{N}}{\partial V} \right) = v \left(\frac{\partial p}{\partial V} \right) \rightarrow -\frac{\bar{\mu}^2}{V} \left(\frac{\partial \bar{N}}{\partial \mu} \right)_{T, V} = v \left(\frac{\partial p}{\partial V} \right)_{T, \bar{N}}$$

$$\hookrightarrow \left(\frac{\partial \bar{N}}{\partial \mu}\right)_{T,V} = - \frac{\bar{N}^2}{U^2} \left(\frac{\partial U}{\partial \mu}\right)_{T,\bar{N}} = \frac{\bar{N}^2}{U} \alpha_T$$

$$\frac{\overline{(\Delta N)^2}}{\bar{N}} = \left(\frac{kT \cdot \alpha_T}{U}\right)^{1/2} \sim \frac{1}{\bar{N}} \sim \frac{1}{N} \quad \text{当 } N, V \rightarrow \infty, \frac{N}{V} = n \rightarrow 0$$

热力学极限.

$$(2) \quad \overline{\Delta E^2} = \overline{E^2} - \bar{E}^2$$

$$= \frac{1}{\Xi} \frac{\partial^2}{\partial \beta^2} \Xi = - \left(\frac{1}{\Xi} \frac{\partial \Xi}{\partial \beta} \right)^2$$

$$= - \left(\frac{\partial \bar{E}}{\partial \beta} \right)_{\alpha, V} = kT^2 \left(\frac{\partial \bar{E}}{\partial T} \right)_{\alpha, V} \quad \beta = \frac{1}{kT}$$

$$= kT^2 \left[\left(\frac{\partial \bar{E}}{\partial V} \right)_{\bar{N}, V} + \left(\frac{\partial \bar{E}}{\partial \bar{N}} \right)_{T, V} \left(\frac{\partial \bar{N}}{\partial T} \right)_{\alpha, V} \right]$$

正则系综的统计力学.

$$= kT^2 \left[C_V + \left(\frac{\partial \bar{E}}{\partial \bar{N}} \right)_{T, V} \left(\frac{\partial \bar{N}}{\partial T} \right)_{\alpha, V} \right] \quad C_V = \left(\frac{\partial \bar{E}}{\partial V} \right)_{\bar{N}, V}$$

$$\left(\frac{\partial \bar{N}}{\partial \beta} \right)_{\alpha, V} = \left(\frac{\partial \bar{E}}{\partial \alpha} \right)_{\beta, V} \Rightarrow \left(\frac{\partial \bar{N}}{\partial T} \right)_{\alpha, V} = \frac{1}{T} \left(\frac{\partial \bar{E}}{\partial \mu} \right)_{T, V}$$

$$= kT^2 C_V + kT \left(\frac{\partial \bar{E}}{\partial \bar{N}} \right)_{T, V} \left(\frac{\partial \bar{E}}{\partial \mu} \right)_{T, V}$$

$$= kT^2 C_V + kT \left(\frac{\partial \bar{E}}{\partial \bar{N}} \right)_{T, V}^2 \cdot \left(\frac{\partial \bar{N}}{\partial \mu} \right)_{T, V}$$

$$= \underbrace{(\Delta E^2)_{CE}}_{\text{正则系综的能量涨落}} + \overline{(\Delta N)^2} \left\{ \left(\frac{\partial \bar{E}}{\partial \bar{N}} \right)_{T, V} \right\}^2 \sim O(N)$$

正则系综的能量涨落.

$$\frac{\overline{(\Delta E)^2}}{\bar{E}} \sim O\left(\frac{1}{N}\right)$$

4. 经典极限 $\lambda_T = \frac{h}{(2\pi mkT)^{1/2}} \ll \delta \bar{r} = n^{-1/3}, \Delta E \ll kT.$

设有 N 个粒子, $S = Nk$ 个自由度.

$$\rho_N(q_1, \dots, p_S) d\Omega_N = \frac{1}{h^S} e^{-\alpha N - \beta \bar{E}_N} d\Omega_N$$

$$d\Omega_N = q_1 - q_5 p_1 - p_5$$

归一化条件: $\sum_N \rho_N d\Omega_N = 1$

§ 与正则系综在例例

例: 单原子分子理想气体的热力学函数.

$$\Xi = \sum_N e^{-\alpha N} Z_N, \quad Z_N = \frac{1}{N! h^{3N}} \int e^{-\beta E_N} d\Omega_N = \frac{z^N}{N!}$$

$$z = \frac{V}{\lambda_T^3} = \frac{V}{h^3} \left(\frac{2\pi m}{\beta} \right)^{3/2}$$

$$\rightarrow \Xi = \sum_N e^{-\alpha N} \frac{z^N}{N!} = \exp(e^{-\alpha} z)$$

$$\ln \Xi = e^{-\alpha} z = \frac{V}{\lambda_T^3} \cdot e^{-\alpha} = \frac{V}{h^3} \left(\frac{2\pi m}{\beta} \right)^{3/2} e^{-\alpha}$$

通常 μ 不可直接测, 通过 $\bar{N}$ 求 μ .

$$\bar{N} = -\frac{\partial}{\partial \alpha} \ln \Xi = \ln \Xi = e^{-\alpha} z.$$

$$\text{得 } \mu = -kT\alpha = kT \ln \left(\frac{\bar{N}}{z} \right) = kT \ln \left(n \lambda_T^3 \right) = kT \ln \left[n h^3 \left(\frac{\beta}{2\pi m} \right)^{3/2} \right]$$

$$\bar{E} = -\frac{\partial}{\partial \beta} \ln \Xi = \frac{3}{2} \frac{1}{\beta} \ln \Xi = \frac{3}{2} \bar{N} kT.$$

$$p = \frac{1}{\beta} \frac{\partial}{\partial V} \ln \Xi = \frac{kT}{V} \bar{N} = n kT.$$

$$S = k \left(\ln \Xi - \alpha \frac{\partial}{\partial \alpha} \ln \Xi - \beta \frac{\partial}{\partial \beta} \ln \Xi \right)$$

$$= k \left(\bar{N} + \alpha \bar{N} + \frac{3}{2} \bar{N} \right) = k \left(\frac{5}{2} + \alpha \right) \bar{N} = k \bar{N} \left(\frac{5}{2} - \frac{\mu}{kT} \right)$$

$$= k \bar{N} \left[\frac{5}{2} - \ln \left(\frac{\bar{N}}{V} \right) - \frac{3}{2} \ln \left(\frac{h^2}{2\pi m kT} \right) + \frac{3}{2} \ln 7 \right]$$

$$= \frac{3}{2} \bar{N} k \ln T + \bar{N} k \ln \left(\frac{V}{N} \right) + \frac{3}{2} \bar{N} k \left\{ \frac{5}{3} + \ln \left[\frac{2\pi m k}{h^2} \right] \right\}$$

例：固体表面的吸附率

假设：固体有 N_0 个吸附中心，每个最多吸附 1 个气态分子，

吸附后分子具有能量 $-\epsilon_0$

定义吸附率： $\theta = \frac{\bar{N}}{N_0}$

外部为单原子分子理想气体，求平衡时 θ 。

$$E_{NS} = -N\epsilon_0$$

$$\Xi = \sum_{N=0}^{N_0} \sum_S e^{-\alpha N - \beta E_{NS}} = \sum_{N=0}^{N_0} \sum_S e^{\beta(\mu + \epsilon_0)N}$$

$$= \sum_{N=0}^{N_0} C_{N_0}^N \cdot e^{\beta(\mu + \epsilon_0)N} = \left(1 + e^{\beta(\mu + \epsilon_0)} \right)^{N_0}$$

$$\ln \Xi = N_0 \ln \left(1 + e^{\beta(\mu + \epsilon_0)} \right)$$

$$\bar{N} = -\frac{\partial}{\partial \alpha} \ln \Xi = \frac{e^{-\alpha + \beta \epsilon_0}}{1 + e^{-\alpha + \beta \epsilon_0}} N_0$$

$$\theta = \frac{\bar{N}}{N_0} = \frac{1}{1 + e^{-\beta(\mu + \epsilon_0)}}$$

其中的 μ 由外部平衡条件给出，

$$\bar{N}' = e^{-\alpha} \bar{z} = e^{-\alpha} \frac{V}{h^3} \left(\frac{2\pi m}{\beta} \right)^{3/2}$$

$$\mu = -kT \alpha = kT \ln \frac{h^3}{(2\pi m kT)^{3/2}}$$

$$= \frac{1}{h^3} \ln \frac{1}{(2\pi m k T)^{3/2} \cdot \frac{1}{kT}}$$

$$\Rightarrow Q = 1 + \frac{kT}{ph^3} (2\pi m k T)^{3/2} e^{-\epsilon_0/kT}$$

$$= P + \frac{1}{P} \frac{2\pi m}{h^3} (kT)^{3/2} e^{-\epsilon_0/kT}$$

$$P \uparrow, Q \uparrow$$

$$T \uparrow, Q \downarrow$$

§ 9 正则系综 $\rightarrow$ Fermi / Bose 分布

结论: 系综分布的平均分布 $\Rightarrow$ 最可几分布

$$N \text{ 粒子分布: } \rho_{N\epsilon} = \frac{1}{Z} e^{-\alpha N - \beta E}, \quad Z = \sum_{N=0}^{\infty} \sum_{\epsilon} e^{-\alpha N - \beta E}$$

把 N, E 用能级分布 $\{a_n\}$ 表达:

$$Z = \sum_{N=0}^{\infty} \sum_{\epsilon} e^{-\alpha N - \beta E_{N\epsilon}} = \sum_{N=0}^{\infty} \sum_{\epsilon_N} \sum_{\epsilon} e^{-\alpha N - \beta E_{N\epsilon}}$$

$\downarrow \quad \downarrow \quad \downarrow$
 $\forall \epsilon_N, s \quad \quad \quad \forall s$

$$= \sum_{N=0}^{\infty} \sum_{\epsilon_N} \sum_{\{a_n\}} W(\{a_n\}) e^{-\alpha \sum_n a_n - \beta \sum_n a_n \epsilon_n}$$

$$= \sum_{\{a_n\}} W(\{a_n\}) e^{-\sum_n (\alpha + \beta \epsilon_n) a_n}$$

$\downarrow$ 此时 $\forall \{a_n\}$ 求和.

$\{a_n\}$ 分布 $\{a_n\}$ 在 $\begin{cases} N = \sum_n a_n & \text{总粒子数} \\ E = \sum_n a_n \epsilon_n & \text{总能量} \end{cases}$

由全同粒子的性质:

$$W(\{a_n\}) = \prod_n W_n = \begin{cases} \frac{g_n!}{a_n! (g_n - a_n)!} & \text{Fermion} \\ \frac{g_n!}{a_n! (g_n - 1)!} & \text{Boson} \end{cases}$$

$$Z = \sum_{\{a_n\}} \left(\prod_n W_n \right) e^{-\sum_n (\alpha + \beta \epsilon_n) a_n}$$

$$= \sum_{\{a_n\}} \prod_n \left[W_n e^{-(\alpha + \beta \epsilon_n) a_n} \right]$$

$\downarrow$ 对所有可能的 $\{a_1, \dots, a_n, \dots\}$ 求和

$$= \sum_{a_1} \sum_{a_2} \dots \sum_{a_N} w_1 e^{-(\alpha + \beta \epsilon_1) a_1} \cdot w_2 e^{-(\alpha + \beta \epsilon_2) a_2} \dots w_N e^{-(\alpha + \beta \epsilon_N) a_N} \dots$$

$$= \left[\sum_{a_1} w_1 e^{-(\alpha + \beta \epsilon_1) a_1} \right] \cdot \dots \cdot \left[\sum_{a_N} w_N e^{-(\alpha + \beta \epsilon_N) a_N} \right] \dots$$

$$= \prod_{\lambda} \sum_{a_{\lambda}} w_{\lambda} e^{-(\alpha + \beta \epsilon_{\lambda}) a_{\lambda}} = \prod_{\lambda} \Xi_{\lambda}$$

$$\Rightarrow \Xi_{\lambda} = \sum_{a_{\lambda}} w_{\lambda} e^{-(\alpha + \beta \epsilon_{\lambda}) a_{\lambda}}$$

$$\textcircled{1} \text{ Fermion } \Xi_{\lambda} = \sum_{a_{\lambda}=0}^{g_{\lambda}} \frac{g_{\lambda}!}{a_{\lambda}! (g_{\lambda} - a_{\lambda})!} e^{-(\alpha + \beta \epsilon_{\lambda}) a_{\lambda}} = \left[1 + e^{-(\alpha + \beta \epsilon_{\lambda})} \right]^{g_{\lambda}}$$

$$\textcircled{2} \text{ Boson } \Xi_{\lambda} = \sum_{a_{\lambda}=0}^{\infty} \frac{(g_{\lambda} + a_{\lambda} - 1)!}{a_{\lambda}! (g_{\lambda} - 1)!} e^{-(\alpha + \beta \epsilon_{\lambda}) a_{\lambda}}$$

$$\begin{aligned} (1-x)^{-m} &= 1 + \frac{(-m)}{1!} (-x) + \frac{(-m)(-m-1)}{2!} (-x)^2 + \dots \\ &\quad + \frac{(-m)(-m-1)\dots(-m-n+1)}{n!} (-x)^n + \dots \end{aligned}$$

$$= \sum_{i=0}^{\infty} \frac{(-1)^i}{i!} \frac{(-1)^i (m+i-1)!}{(m-1)!} x^i = \sum_{i=0}^{\infty} \frac{(m+i-1)!}{i! (m-1)!} x^i$$

$$\Xi_{\lambda} = \left[1 - e^{-(\alpha + \beta \epsilon_{\lambda})} \right]^{-g_{\lambda}}$$

$$\Rightarrow \ln \Xi = \pm \sum_{\lambda} g_{\lambda} \ln \left[1 \pm e^{-(\alpha + \beta \epsilon_{\lambda})} \right] \begin{cases} + \text{ for Fermion} \\ - \text{ for Boson} \end{cases}$$

$$\bar{a}_{\mu} = \sum_{\lambda} \sum_s a_{\lambda} \rho_{\mu s}$$

$$= \frac{1}{\Xi} \sum_{\lambda} \sum_s \sum_{\{a_{\lambda}\}} a_{\lambda} w(\{a_{\lambda}\}) e^{-\sum_{\lambda} (\alpha + \beta \epsilon_{\lambda}) a_{\lambda}}$$

$$= \frac{1}{\Xi} \sum_{\{a_{\lambda}\}} a_{\lambda} \prod_{\lambda} \left[w_{\lambda} e^{-(\alpha + \beta \epsilon_{\lambda}) a_{\lambda}} \right]$$

$$= \frac{\sum_{\lambda \neq K} w_{\lambda} e^{-(\alpha + \beta \epsilon_{\lambda}) a_{\lambda}}}{\prod_{\lambda} w_{\lambda} e^{-(\alpha + \beta \epsilon_{\lambda}) a_{\lambda}}} \sum_{a_K} a_K w_K e^{-(\alpha + \beta \epsilon_K) a_K}$$

$$= \sum_{a_K} a_K w_K e^{-(\alpha + \beta \epsilon_K) a_K} \quad \Xi_K = \sum_{a_K} w_K e^{-(\alpha + \beta \epsilon_K) a_K}$$

$$\sum_{a_K} w_K e^{-(\alpha + \beta \epsilon_K) a_K} = - \frac{\partial}{\partial \alpha} \ln \Xi_K$$

$$g_K, + \text{ for fermion}$$

2

$$e^{\alpha + \beta \epsilon_k} \neq 1$$

- for Bose: